

WASP-11b

R-0695

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_250209 · created 2026-07-27T21:04:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460715.6493 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.051
Transit depth (Rp/Rs)^2	2.35 +/- 1.55 [%]
Orbital Inclination (inc)	84.75 +/- 1.74
Airmass coefficient 1 (a1)	5.39 +/- 3.04
Airmass coefficient 2 (a2)	-0.83 +/- 0.43
Scatter in the residuals of the lightcurve fit is	54.55 %
Best Comparison Star	None
Optimal Aperture	2.17
Optimal Annulus	3.0
Transit Duration (day)	0.035 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.153 $\pm$ 0.051 vs 0.131 (deviation 0.35 σ)
Duration fit vs published	0.035 d vs 0.1075 d (deviation 1.6 σ)
Mid-time O-C	-6.8 min
Depth SNR	2.5
Residual scatter	54.55 %

Light curve

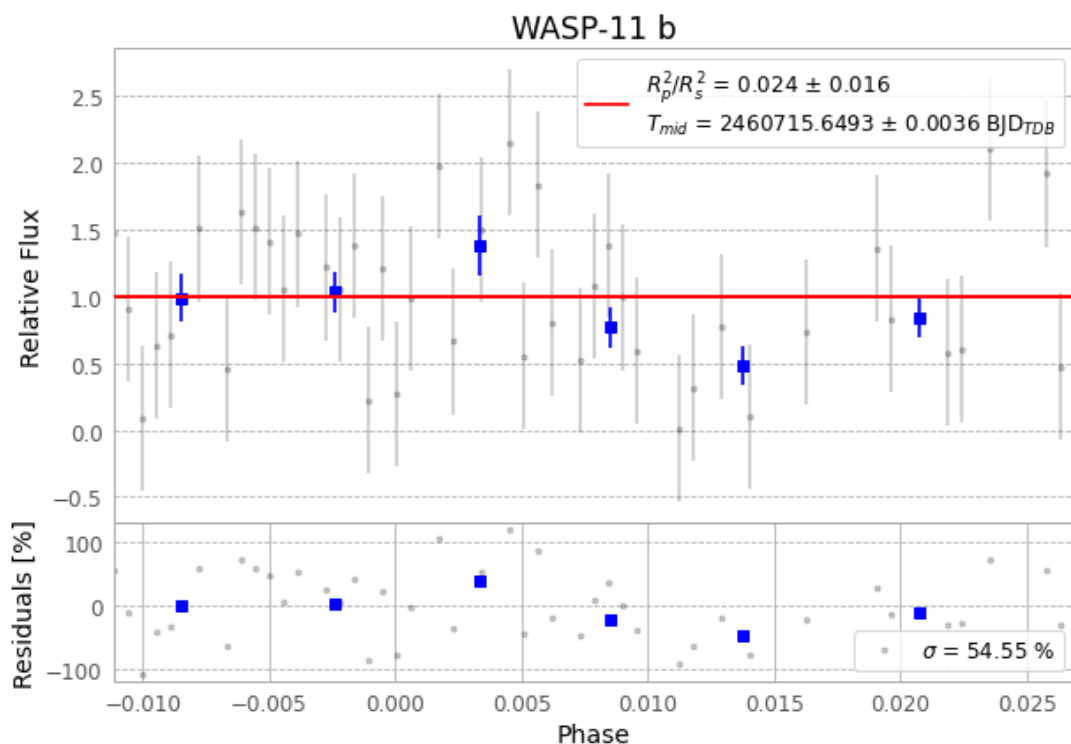


Figure 1 — FinalLightCurve_WASP-11 b_2025-02-08.png (SHA-256 5478cff19b8cb481...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [374, 366], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 142423cbeb6eec4a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

71 FITS frames (47 MB), HATP-10250209023015.FITS → HATP-10250209060016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-250209.log (ca841824bd1f...), FinalLightCurve_WASP-11 b_2025-02-08.png (5478cff19b8c...), FinalLightCurve_WASP-11 b_2025-02-08.csv (8f64f1bbdaa3...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 21:04Z.